



HOSPITAL MANAGEMENT SYSTEM

DIPTARKA BISWAS, 12 S3 25
BOARD ROLL NO: 12677140



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ACKNOWLEDGEMENT

It is with pleasure that I acknowledge my sincere gratitude to my teacher Mr. Sujit Chatterjee and Mr. Ranjit Mondal who both taught and undertook the responsibility of teaching the subject of Computer Science. I have greatly benefitted from their classes.

I am especially indebted to our principal, Mr. Arun Kumar Dasgupta who has always been a source of encouragement and support and without whose inspiration this project would not have been a successful endeavor. I would like to place on record heartfelt thanks to him.

Finally, I would like to express my sincere appreciation to all the other students of my batch for their friendship and constant support.



CERTIFICATE

This is to certify that

DIPTARKA BISWAS

of

BHAVAN'S GANGABUX KANORIA VIDYAMANDIR

has done the project on

“HOSPITAL MANAGEMENT SYSTEM”

under my supervision. He has taken interest and has shown utmost sincerity in the completion of the project.

I certify this project up to my expectation as per guidelines issued by CBSE, New Delhi.

Teacher's Signature

Internal Examiner

External Examiner

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware:

1. Personal Computer
2. Internet Connection(for setting up software)

Software:

1. Python 3.14(or higher)
2. MySQL v8.0.39(or higher)
3. MySQL – Python Connector Module





HOSPITAL MANAGEMENT SYSTEM

A hospital management system (HMS) is an integrated software solution designed to streamline healthcare operations by combining **clinical**, **administrative**, and **financial** workflows into one platform. It helps hospitals manage **patient records**, **appointments**, **billing**, **pharmacy inventory** with accuracy and efficiency. By **digitizing** processes, HMS reduces **paperwork**, **minimizes errors**, and ensures **faster access** to patient information. For administrators, HMS provides tools for **resource allocation**, **reporting**, and **decision-making**, while for medical staff it enables **quick access** to **patient histories and treatment plans**. Patients benefit from **smoother registration**, **reduced waiting times**, and **transparent billing**. Overall, a hospital management system enhances productivity, ensures data security, and delivers better patient care by creating a centralized, user-friendly environment for managing complex healthcare operations.

► Early History of HMS

- 1960~: HMS began as basic financial systems in hospitals, held back by expensive mainframe hardware and lack of networking.
- El Camino Hospital, California was one of the first to experiment with computerized systems in the late 1960s.
- 1970–1980: As personal computers and LANs came into being, hospitals started integrating clinical and administrative modules.

- 
- 1980~: Introduction of WANs allowed data sharing across hospital sites, laying the groundwork for modern HMS.

► Institutional Milestones

- HIMSS (Healthcare Information and Management Systems Society) was founded in 1961 to support the development and adoption of HMS technologies. It later evolved into the Healthcare Information and Management Systems Society.
- Early systems were often custom-built for large hospitals and focused on billing, scheduling and patient records.

NEED OF HMS

- Centralized platform: Integrates clinical, administrative, and financial workflows.
- Patient management: Handles records, appointments, and treatment histories.
- Pharmacy & billing: Tracks inventory and ensures accurate billing.
- Efficiency: Reduces paperwork, minimizes errors, and speeds up access to information.
- For administrators: Provides tools for resource allocation, reporting, and decision-making.
- For medical staff: Enables quick access to patient histories and treatment plans.
- For patients: Improves registration, reduces waiting times, and ensures transparent billing.
- Data security: Protects sensitive healthcare information.



FUTURE SCOPE

Future scopes of this hospital management system include AI-driven diagnostics, predictive analytics for patient care, and seamless integration with wearable health devices. Cloud-based HMS will enable remote access. These innovations aim to personalize treatment, improve efficiency, and support proactive healthcare delivery.

---CODE---

Setting up MySQL-Python Connector:

---Prerequisites:

1. Ensure pip is configured. If not, reinstall python with pip configured.
2. Python is on system's PATH. If not, reinstall. The option to add to PATH is usually available during installation.
3. Administrator privileges.

---To install:

1. Open any windows command shell
2. Type:
cd "<python directory path>"
pip install mysql-connector-python

Setting up Tabulate module(for formatted outputs):

1. Open any windows command shell
2. Type:
cd "<python directory path>"
pip install tabulate



=PROGRAM=

Program Outputs:

Admin menu:

```
PS E:\Education\Computer Science\Python\Hospital Management Class 12 Project\hms-test-v1.1> python project_main.py
Ensuring database exists...
Default admin 'superuser000' created (password 'admin123457').
Database initialization complete.
Welcome to the Hospital-Management-System (v1.0)
=== MENU ===
1. Login
2. Register as Patient
3. Exit
Choice: 1
=== LOGIN ===
Username: superuser000
Password: admin123457
Logged in as default_admin_0,admin
=== ADMIN MENU ===
1. Create employee (doctor/pharmacist)
2. Remove employee
3. Modify pharmacy stock (add/update)
4. View pharmacy stock
5. View sales log
6. Add beds
7. Remove bed
8. View beds
9. View all prescriptions
10. Change account password
0. Logout
```

Add Beds:

```
=== ADMIN MENU ===
1. Create employee (doctor/pharmacist)
2. Remove employee
3. Modify pharmacy stock (add/update)
4. View pharmacy stock
5. View sales log
6. Add beds
7. Remove bed
8. View beds
9. View all prescriptions
10. Change account password
0. Logout
Choice: 6
How many beds to add (integer): 100
Added 100 beds.
```

View Beds:

```
=== ADMIN MENU ===
1. Create employee (doctor/pharmacist)
2. Remove employee
3. Modify pharmacy stock (add/update)
4. View pharmacy stock
5. View sales log
6. Add beds
7. Remove bed
8. View beds
9. View all prescriptions
10. Change account password
0. Logout
Choice: 8
+-----+-----+-----+
| bed_id | status | patient_uid |
+-----+-----+-----+
| 1 | Available | |
+-----+-----+-----+
| 2 | Available | |
+-----+-----+-----+
| 3 | Available | |
+-----+-----+-----+
| 4 | Available | |
+-----+-----+-----+
| 5 | Available | |
+-----+-----+-----+
```

Create Employee Record:

```
=== ADMIN MENU ===
1. Create employee (doctor/pharmacist)
2. Remove employee
3. Modify pharmacy stock (add/update)
4. View pharmacy stock
5. View sales log
6. Add beds
7. Remove bed
8. View beds
9. View all prescriptions
10. Change account password
0. Logout
Choice: 1
Create new employee (doctor or pharmacist)
Name: Dr. Doctor
Username: def_doc_0
Password: 1234
Role (1. doctor/ 2. pharmacist): 1
1. General Medicine
2. Cardiology
3. Neurology
4. Orthopedics
5. Pediatrics
6. Dermatology
7. ENT
8. Gynecology
9. Nephrology
Choose: 1
Employee created with uid 2
3. View past prescriptions for a patient
4. Assign bed to patient
5. Release bed (discharge)
6. Change account password
0. Logout
Choice: 2
Patient uid: 8
Patient found: Patient0, uid: 8
Enter prescription text. End with a blank line.
Fever
--Paracetamol 500 (max 3 times daily)
--Azithromycin (Twice daily, 5 DAYS)

Patient name: Mrs. Patient
Patient age: 25
Patient gender: F
Patient phone: 9876543210
Disease / Diagnosis: Fever
Prescription created (id 1).
```

Doctor Menu:

```
=== LOGIN ===
Username: def_doc_0
Password: 1234
Logged in as Dr. Doctor,doctor
=== DOCTOR MENU ===
1. View patient list
2. Create prescription
3. View past prescriptions for a patient
4. Assign bed to patient
5. Release bed (discharge)
6. Change account password
0. Logout
Choice: 1
```

Doctor Functions:

```
=== DOCTOR MENU ===
1. View patient list
2. Create prescription
3. View past prescriptions for a patient
4. Assign bed to patient
5. Release bed (discharge)
6. Change account password
0. Logout
Choice: 1
+-----+
| uid | name |
+-----+
| 8 | Patient0 |
+-----+
| 9 | Patient1 |
+-----+
| 10 | Patient2 |
+-----+
=== DOCTOR MENU ===
1. View patient list
2. Create prescription
3. View past prescriptions for a patient
4. Assign bed to patient
5. Release bed (discharge)
6. Change account password
0. Logout
Choice: 2
Patient uid: 8
Patient found: Patient0, uid: 8
Enter prescription text. End with a blank line.
Fever
--Paracetamol 650 (max 3 times daily)
--Azithromycin (Twice daily, 5 DAYS)

Patient name: Mrs. Patient
Patient age: 25
Patient gender: F
Patient phone: 9876543210
Disease / Diagnosis: Fever
Prescription created (id 1).
```

```
Choice: 3
Patient uid to view: 8
+-----+-----+-----+
| prescription_id | doctor | description | date_issued |
+-----+-----+-----+
| 1 | Dr. Doctor 2 | Fever | 2025-12-02 11:59:42 |
| | | --Paracetamol 650 (max 3 times daily) | |
| | | --Azithromycin (Twice daily, 5 DAYS) | |
+-----+-----+-----+
```

Patient Registration:

```
=== PATIENT REGISTRATION ===
Name: Patient0
Choose a username: def_pat_0
Choose a password: 1234
Patient registered successfully.
```

Pharmacy:

```
=== ADMIN MENU ===
1. Create employee (doctor/pharmacist)
2. Remove employee
3. Modify pharmacy stock (add/update)
4. View pharmacy stock
5. View sales log
6. Add beds
7. Remove bed
8. View beds
9. View all prescriptions
10. Change account password
0. Logout
Choice: 3
Modify stock (add new med or update existing)
1. New
2. Update
Choice: 1
Medicine name: Parasafe 650
Quantity (integer): 100
Price per unit: 28.50
Medicine added.
```

=== ADMIN MENU ===

1. Create employee (doctor/pharmacist)
 2. Remove employee
 3. Modify pharmacy stock (add/update)
 4. View pharmacy stock
 5. View sales log
 6. Add beds
 7. Remove bed
 8. View beds
 9. View all prescriptions
 10. Change account password
 0. Logout
- Choice: 4

med_id	med_name	quantity	price
2	Clavam 625	100	195
3	02	35	175.1
4	Pan 40	200	150
1	Parasafe 650	100	28.5

=== PHARMACIST MENU ===

1. View stock
 2. Update stock
 3. Sell
 4. View sales log
 5. Change account password
 0. Logout
- Choice: 3
- Enter med_id to add to cart (or blank to checkout): 4
Selected: {'med_id': 4, 'med_name': 'Pan 40', 'quantity': 200, 'price': Decimal('150.00')}
- Quantity: 2
Added to cart.
- Enter med_id to add to cart (or blank to checkout): 3
Selected: {'med_id': 3, 'med_name': '02', 'quantity': 35, 'price': Decimal('175.10')}
- Quantity: 1
Added to cart.
- Enter med_id to add to cart (or blank to checkout):
Confirm checkout? (y/n): y

=== BILL ===

Med ID	Name	Qty	Unit Price	Line Total
4	Pan 40	2	150	300
3	02	1	175.1	175.1

TOTAL: 475.1

Proceed to checkout? (y/n): y

Checkout successful. Total: 475.1

=== PHARMACIST MENU ===

1. View stock
 2. Update stock
 3. Sell
 4. View sales log
 5. Change account password
 0. Logout
- Choice: 4

sale_id	med_name	quantity_sold	total_price	sold_by	sold_at
2	Pan 40	2	300	Mr. Pharmacist	2025-12-02 12:26:25
3	02	1	175.1	Mr. Pharmacist	2025-12-02 12:26:25
1	Parasafe 650	3	85.5	Mr. Pharmacist	2025-12-02 12:25:50

SQL TABLES:

Beds:

```
mysql> select * from beds;
```

bed_id	status	patient_uid
1	Occupied	10
2	Occupied	8
3	Available	NULL
4	Available	NULL
5	Available	NULL
6	Available	NULL
7	Available	NULL
8	Available	NULL
9	Available	NULL
10	Available	NULL
11	Available	NULL
12	Available	NULL
13	Available	NULL

Users:

```
100 rows in set (0.00 sec)

mysql> select * from users;
+-----+-----+-----+-----+-----+-----+
| uid | username | password | name | role | specialization |
+-----+-----+-----+-----+-----+-----+
| 1 | superuser000 | admin123457 | default_admin_0 | admin | NULL |
| 2 | def_doc_0 | 1324 | Dr. Doctor 0 | doctor | General Medicine |
| 3 | def_doc_1 | 1234 | Dr. Doctor 1 | doctor | Cardiology |
| 4 | def_doc_3 | 1234 | Dr. Doctor 2 | doctor | Neurology |
| 5 | def_doc_4 | 1234 | Dr. Doctor 4 | doctor | ENT |
| 6 | def_pha_1 | 1234 | Mr. Pharmacist | pharmacist | NULL |
| 7 | def_pha_2 | 1234 | Mr. Pharmacist 2 | pharmacist | NULL |
| 8 | def_pat_0 | 1234 | Patient0 | patient | NULL |
| 9 | def_pat_1 | 1234 | Patient1 | patient | NULL |
| 10 | def_pat_2 | 0000 | Patient2 | patient | NULL |
| 11 | def_patient_4 | 1234 | Patient4 | patient | NULL |
+-----+-----+-----+-----+-----+-----+

11 rows in set (0.00 sec)
```

Prescriptions:

```
mysql> select prescription_id as pid, doctor_uid as d_id, patient_uid as p_id, patient_name as name, patient_gender as gen, patient_phone as phone, description as desc, disease from prescriptions;
+-----+-----+-----+-----+-----+-----+-----+
| pid | d_id | p_id | name | gen | phone | desc |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | 3 | 8 | Mrs. Patient | F | 9876543210 | Fever |
--Paracetamol 650 (max 3 times daily) |
--Azithromycin (Twice daily, 5 DAYS) |
| 2 | 2 | 10 | Mr. Patient | M | 9874561320 | Stomach Pain |
Diarrhea |
Gas |
---Pantoprazole 40 (On empty stomach in the morning until symptoms subside.) |
---02(Antibiotic) (Twice daily, 5days) | Stomach Infection |
| 3 | 5 | 10 | Mrs. Patient | F | 7854692163 | Thyroid Cancer. Immediate Surgery recommended. |
| Thyroid Cancer |
| 4 | 4 | 11 | Mr. Patient | M | 8791534625 | Migraine |
---Flunarilin 10(With onset of symptoms, take 1 tablet) |
---Avoid loud noises and bright lights | Migraine |
+-----+-----+-----+-----+-----+-----+-----+

4 rows in set (0.00 sec)
```

(formatting problems due to too long description text)

Pharmacy:

```
mysql> select * from pharmacy;
+-----+-----+-----+-----+
| med_id | med_name | quantity | price |
+-----+-----+-----+-----+
| 1 | Parasaft 650 | 97 | 28.50 |
| 2 | Clavam 625 | 100 | 195.00 |
| 3 | 02 | 34 | 175.10 |
| 4 | Pan 40 | 198 | 150.00 |
+-----+-----+-----+-----+

4 rows in set (0.00 sec)
```

Sales:

```
mysql> select * from sales_log;
+-----+-----+-----+-----+-----+-----+
| sale_id | med_id | quantity_sold | total_price | sold_by | sold_at |
+-----+-----+-----+-----+-----+-----+
| 1 | 1 | 3 | 85.50 | 6 | 2025-12-02 12:25:50 |
| 2 | 4 | 2 | 390.00 | 6 | 2025-12-02 12:26:25 |
| 3 | 3 | 1 | 175.10 | 6 | 2025-12-02 12:26:25 |
+-----+-----+-----+-----+-----+-----+

3 rows in set (0.00 sec)
```



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